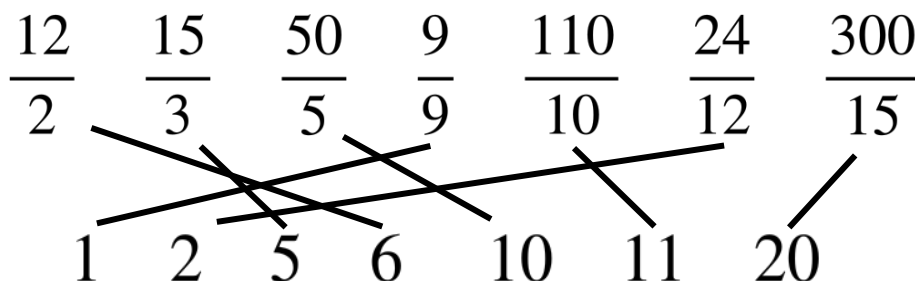




Checking and securing converting mixed numbers to improper fractions

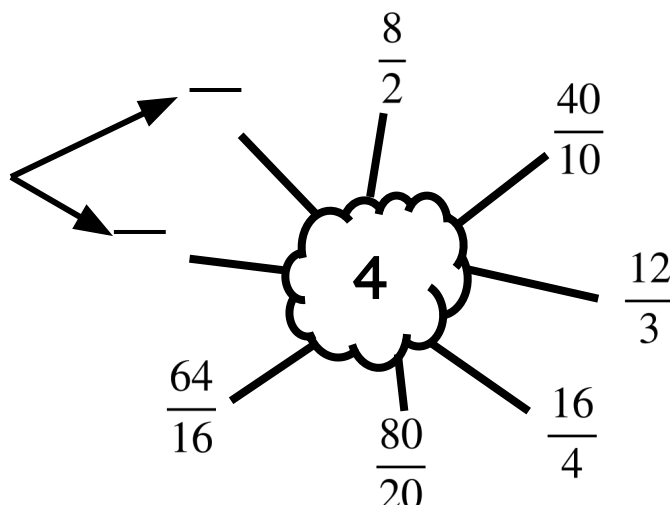
Task A: Integers as improper fractions

1) Match the improper fraction with the correct integer.

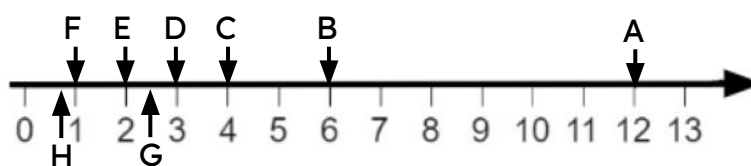


2) Fill in the gaps to evaluate the answer to 4.

There are an infinite number of improper fractions which evaluate to 4



3) Place the following on the number line.



a) $\frac{12}{1}$

b) $\frac{12}{2}$

c) $\frac{12}{3}$

d) $\frac{12}{4}$

e) $\frac{12}{6}$

f) $\frac{12}{12}$

g) Estimate where $\frac{12}{5}$ would lie.

h) Estimate where $\frac{12}{13}$ would lie.



4) The , ,  and  are either

3	4	12	24
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$$\frac{\text{cat face}}{\text{flower}} + \frac{\text{star}}{\text{flower}} = 9$$

$$\frac{\text{star}}{\text{flower}} - \frac{\text{cat face}}{\text{dog face}} = 2$$

$$\frac{\text{star}}{\text{cat face}} + \frac{\text{dog face}}{\text{dog face}} = \text{dog face}$$

To make the above calculations correct, work out the value of the images.

$$\text{cat face} = 12$$

$$\text{flower} = 4$$

$$\text{dog face} = 3$$

$$\text{star} = 24$$

Task B: Mixed numbers to improper fractions

1) Fill in the blanks to show the working out and conversion from a mixed number to an improper fraction.

$$\begin{aligned} \text{a) } 15 \frac{3}{4} &= 15 \times \frac{4}{4} + \frac{3}{4} \\ &= \frac{60}{4} + \frac{3}{4} \\ &= \frac{63}{4} \end{aligned}$$

$$\begin{aligned} \text{b) } 7 \frac{2}{9} &= 7 \times \frac{9}{9} + \frac{2}{9} \\ &= \frac{63}{9} + \frac{2}{9} \\ &= \frac{65}{9} \end{aligned}$$

2) Convert the following to improper fractions, showing working out if you need.

$$\text{a) } 4 \frac{1}{2} = \frac{9}{2}$$

$$\text{b) } 11 \frac{3}{5} = \frac{58}{5}$$

$$\text{c) } 80 \frac{1}{3} = \frac{241}{3}$$

$$\text{d) } 3 \frac{9}{10} = \frac{39}{10}$$



3) Can you work out the mixed number given the clues for the integer and proper fraction?

$$10\frac{3}{5}$$

The integer is even.
The numerator and denominator of the proper fraction are both odd and prime numbers.

between $\frac{46}{5}$ and 10.7

$$24\frac{5}{7}$$

The integer is even.
As an improper fraction the numerator is 173.
The denominator is odd.

The mixed number is
between $\frac{124}{5}$ and $\frac{116}{5}$

Write your own mixed number with clues.

Task C: Comparing mixed numbers and improper fractions

1) Insert the symbol $<$, $>$ or $=$ to make the following statements true.

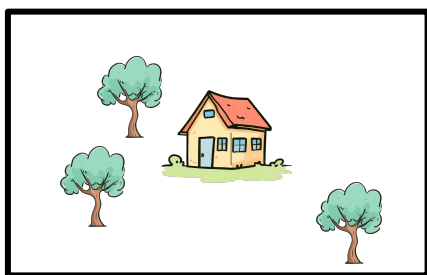
a) $\frac{13}{4} = 3\frac{1}{4}$

b) $\frac{24}{5} < 5\frac{1}{5}$

c) $\frac{92}{4} > 22\frac{1}{4}$

d) $\frac{9}{2} > \frac{25}{6}$

$$4\frac{1}{5}\text{ m}$$



$$\frac{49}{5}\text{ m}$$

2) Two gardeners are putting a fence around a rectangular garden.

One gardener measures the length to be $\frac{49}{5}\text{ m}$ and the other gardener measures the width to be $4\frac{1}{5}\text{ m}$.

Fence panels cost £22 per m.

a) How many fence panels are needed?

$$4\frac{1}{5} + \frac{49}{5} + 4\frac{1}{5} + \frac{49}{5} =$$

$$\frac{21}{5} + \frac{49}{5} + \frac{21}{5} + \frac{49}{5} = \frac{140}{5} = 28 \text{ fence panels}$$

b) What is the total cost?

$$28 \times 22 = £616$$



3) The following have been put in ascending order.

The mixed number is above and the equivalent improper fraction is below.

Fill in the gaps.

a) $1\frac{1}{2}$, $2\frac{3}{4}$, $5\frac{1}{10}$, $5\frac{3}{4}$, $20\frac{1}{2}$
 $\frac{3}{2}$, $\frac{11}{4}$, $\frac{51}{10}$, $\frac{23}{4}$, $\frac{41}{2}$

b) $2\frac{2}{3}$, $4\frac{4}{5}$, $4\frac{9}{10}$, $5\frac{1}{3}$, $5\frac{2}{5}$
 $\frac{8}{3}$, $\frac{24}{5}$, $\frac{49}{10}$, $\frac{16}{3}$, $\frac{27}{5}$